

Humaun Kabir

Ph.D. Researcher · Materials Science & Engineering · Texas A&M University

mkhumaun@tamu.edu · (979) 255-1245 · [Google Scholar ↗](#) · [ORCID ↗](#) · [LinkedIn ↗](#)

RESEARCH PROFILE

Materials scientist integrating experiments, density functional theory, and molecular dynamics to engineer functional materials and fluids for lubrication, thermal management, and electrified systems. Research links interfacial chemistry with friction, wear, thermal transport and field-responsive behavior.

EDUCATION

Ph.D., Materials Science & Engineering 2023–present

Texas A&M University

Advisor: Hong Liang · Expected completion: December 2027

Graduate study toward M.Phil., Solid State Physics 2020–2022

Chittagong University of Engineering and Technology

36 of 48 credits completed

B.Sc., Materials & Metallurgical Engineering 2015–2019

Bangladesh University of Engineering and Technology

RESEARCH & TEACHING EXPERIENCE

Graduate Research Assistant 2023–present

Texas A&M University · Surface Science Research Group

J. Mike Walker '66 Department of Mechanical Engineering · PI: Hong Liang

Surface-engineered lubrication. Developed functionalized MoS₂ with 30-day dispersion stability in water, EG and glycerol. Reduced wear by 93% in PAO and 81% in water versus base fluids, and 85% in lithium grease versus unmodified MoS₂. DDA-CNPs achieved over 60-day PAO suspension stability and 26% lower friction at 1 N versus base oil.

Electrified interfaces. Investigated MoS₂- and carbon-modified greases, achieving 93% lower wear with MoS₂ at 40 °C and up to approximately 45% lower wear with CNPs, both at 3 A versus corresponding base greases. Distinguished bulk impedance from contact resistance; proposed an electrified-damage framework in an Editor's Choice Perspective.

Electro-responsive fluids. Engineered CNPs@TiO₂ suspensions with approximately 1,870% viscosity enhancement over the off-field baseline without voltage collapse. APTS–SiO₂ achieved approximately 4.9 times the yield stress of bare silica, both at 3 wt.% and 3 kV/mm.

Thermal-fluid design. Combined 22 water–ethylene glycol–glycerol formulations and six DFT dimer models to develop a cohesion descriptor and identify a cooling–lubrication formulation window; manuscript submitted. Ongoing LAMMPS nonequilibrium MD benchmarks mixture thermal conductivity against experiments.

Lecturer Jan 2020–Jan 2023

Chittagong University of Engineering and Technology

Teaching and research in materials science.

Research Assistant May–Nov 2019

Bangladesh University of Engineering and Technology

Department of Materials and Metallurgical Engineering.

TECHNICAL EXPERTISE

Synthesis & surface engineering: Hydrothermal and sol–gel synthesis, spin coating, controlled oxidation, amine and silane functionalization, particle dispersion and formulation.

Thermophysical & tribological testing: Thermal conductivity, rheometry, friction and wear, adhesion, EIS, contact resistance, electrorheology and leakage-current measurements.

Materials characterization: SEM/EDS, XRD, XPS, FTIR, UV–Vis, TGA, 3D profilometry and wettability.

Computational modeling: Quantum ESPRESSO, CASTEP, LAMMPS, ABCluster and xTB; optimization, interfacial energetics, electronic and elastic properties, electrostatics and eHEX thermal transport.

PATENTS & INTELLECTUAL PROPERTY

Liang, H.; Kabir, M. H. Functionalized-Nanoparticle Enabled Metal Working Fluids and Coolants. U.S. Non-Provisional Patent Application No. **19/798,316**, filed 4 September 2026. Claims priority to U.S. Provisional Application No. 63/878,394.

HONORS & AWARDS

NLGI Student Poster Board Award & Travel Grant NLGI Annual Meeting	2026
STLE Houston Section Scholarships Society of Tribologists and Lubrication Engineers	2026 · 2024
Dean's List Award Final undergraduate year	BUET
Merit Scholarship Final undergraduate year; top 10% of students	BUET
University Stipend Scholarship Five semesters	BUET

SELECTED CONFERENCE PRESENTATIONS

Kabir, M. H.; Tajedini, M.; Liang, H. Designing Conductive Pathways in Lubricating Grease to Reduce Electrical Pitting in Bearings. *NLGI Annual Meeting*, June 2026.

Kabir, M. H.; Dias, D.; Liang, H. Urea-enabled Suspension of MoS₂ in Water-based Lubricants. *80th STLE Annual Meeting & Exhibition*, May 2026.

Kabir, M. H.; Arole, K.; Green, M. J.; Liang, H. Engineering MXene Layers for Tunable Electrical and Frictional Performance. *80th STLE Annual Meeting & Exhibition*, May 2026.

Kabir, M. H.; Dias, D.; Liang, H. Suspension-ability and Tribological Performance of Functionalized Carbon Nanoparticles in PAO Oil. *79th STLE Annual Meeting & Exhibition*, May 2025.

Kabir, M. H.; Dias, D.; Arole, K.; Liang, H. Urea-functionalized hydrophilic MoS₂ for Grease. *78th STLE Annual Meeting & Exhibition*, May 2024.

SUBMITTED MANUSCRIPT

Kabir, M. H.; Dias, D.; Bons, J.; Liang, H. Hydrogen-Bond Cohesion Governed Thermal and Tribological Performance of Water–Ethylene Glycol–Glycerol. **Manuscript submitted.**

PUBLICATIONS

Complete supplied record: 27 published works. † Authors contributed equally. Cover features and Editor's Choice are annotations, not additional publications.

First- and co-first-author journal articles

1. **Kabir, M. H.**; Dias, D.; Bons, J.; Foysal, M. W.; Liang, H. APTS-SiO₂ enabled field-induced rapid assembly in low-loading suspensions. *Applied Surface Science* 2027, 751, 168138. 10.1016/j.apsusc.2026.168138. Published online 17 August 2026.
2. **Kabir, M. H.**; Dias, D.; Bons, J.; Foysal, M. W.; Johnson, E.; Kosmoski, J.; Liang, H. TiO₂-modified Carbon Nanoparticles (CNPs@TiO₂) Enabled Interfacial Charge Regulation. *ACS Applied Materials & Interfaces* 2026, 18, 42189–42203. 10.1021/acsami.6c10772. [Cover feature]
3. Li, Y.-S.†; **Kabir, M. H.**†; Dias, D.; Bons, J.; Presswood, R.; Cummins, J.; Liang, H. Renewable Carbon Nano-onions Induced Friction and Wear Reduction in Ca-Grease. *ACS Applied Engineering Materials* 2026, 4, 2634–2643. 10.1021/acsaenm.6c00302. [Equal-contribution co-first author]
4. **Kabir, M. H.**; Dias, D.; Bons, J.; Foysal, M. W.; Johnson, E.; Kosmoski, J.; Liang, H. Influence of Carbon Nanoparticles on Electrically Induced Wear of Grease-Lubricated Steels. *Advanced Materials Interfaces* 2026, 13, e70520. 10.1002/admi.70520. [Cover feature]
5. **Kabir, M. H.**; Dias, D.; Presswood, R. G., Jr.; Liang, H. Effects of Waste-Plastic-Derived Carbon Nano-Onions as Additive on Rheological and Tribological Performance of Oil Lubricant. *Journal of Tribology* 2026, 148, 032102. 10.1115/1.4069957.
6. **Kabir, M. H.**; Dias, D.; Velhal, M.; Liang, H. Effects of oxidation-assisted grafting of MoS₂ on tribological performance of steels. *Tribology International* 2025, 211, 110826. 10.1016/j.triboint.2025.110826.
7. **Kabir, M. H.**; Dias, D.; Marjuban, S. M. H.; Avais, M.; Castaneda, H.; Liang, H. Effects of urea-functionalized MoS₂ on hydrophilic lubrication. *Tribology International* 2025, 203, 110384. 10.1016/j.triboint.2024.110384.
8. **Kabir, M. H.**; Dias, D.; Johnson, E.; Kosmoski, J.; Liang, H. Non-Polar Chain-Enabled Suspension of Carbon Nanoparticles in Base Oil. *Lubricants* 2024, 12, 373. 10.3390/lubricants12110373.
9. **Kabir, M. H.**; Dias, D.; Arole, K.; Bahrami, R.; Sue, H.-J.; Liang, H. Hydrophilized MoS₂ as Lubricant Additive. *Lubricants* 2024, 12, 80. 10.3390/lubricants12030080. [Cover feature]
10. **Kabir, M. H.**; Hossain, M. Z.; Jalil, M. A.; Ghosh, S.; Hossain, M. M.; Ali, M. A.; Khandaker, M. U.; Jana, D.; Rahman, M. M.; Hossain, M. K.; Chowdhury, J.; Kazi, M.; Uddin, M. M. The efficacy of rare-earth-doped V₂O₅ photocatalyst for removal of pollutants from industrial wastewater. *Optical Materials* 2024, 147, 114724. 10.1016/j.optmat.2023.114724.
11. **Kabir, M. H.**; Hossain, M. M.; Ali, M. A.; Uddin, M. M.; Ali, M. L.; Hasan, M. Z.; Islam, A. K. M. A.; Naqib, S. H. First-principles study of mechanical, thermal, electronic, optical and superconducting properties of C40-type germanide-based MGe₂ (M = V, Nb and Ta). *Results in Physics* 2023, 51, 106701. 10.1016/j.rinp.2023.106701.
12. **Kabir, M. H.**; Ibrahim, H.; Ayon, S. A.; Billah, M. M.; Neaz, S. Structural, nonlinear optical and antimicrobial properties of sol-gel derived, Fe-doped CuO thin films. *Heliyon* 2022, 8, e10609. 10.1016/j.heliyon.2022.e10609.

Perspective

13. **Kabir, M. H.**; Nguyen, A.; Liang, H. Electrified Damage in Motion Systems. *Advanced Engineering Materials* 2026, 28, e71036. 10.1002/adem.71036. [Perspective · Editor's Choice]

Co-authored journal articles

14. Velhal, M.; Sattari, R.; **Kabir, M. H.**; Li, Y.-S.; Liang, H. Role of Interlayer Expansion and Surface Chelation in Ternary NiCuAl Layered Double Hydroxides for Ultrahigh Capacity Dye Adsorption. *Global Challenges* 2026, 10, e70149. 10.1002/gch2.70149.

15. Bons, J.; **Kabir, M. H.**; Dias, D.; Chertakovsky, R.; Liang, H. Electrified performance characteristics of greases and their effects on wear mechanisms of steels. *Wear* 2026, 598, 206774. 10.1016/j.wear.2026.206774.
16. Tajedini, M.; Fang, L.; Bons, J.; **Kabir, M. H.**; Johnson, E.; Kosmoski, J.; Liang, H. Effects of Carbon Nanoparticles on Triboelectro-Induced Surface Defects. *Advanced Engineering Materials* 2026, 28, e202502084. 10.1002/adem.202502084.
17. Tajedini, M.; **Kabir, M. H.**; Azhdari, R.; Bahrami, R.; Sue, H.-J.; Liang, H. MoS₂ Additives in Lithium Grease for Electrified Systems. *Lubricants* 2025, 13, 106. 10.3390/lubricants13030106.
18. Arole, K.; Pas, S. E.; Thakur, R. M.; Amiouny, L. A.; **Kabir, M. H.**; Dujovic, M.; Radovic, M.; Lutkenhaus, J. L.; Green, M. J.; Liang, H. Effects of Intercalation on ML-Ti₃C₂T_z MXene Properties and Friction Performance. *ACS Applied Materials & Interfaces* 2024, 16, 64156–64165. 10.1021/acsami.4c12659.
19. Dujana, W. A.; Ahmad, S.; Noman, M. N. H.; **Kabir, M. H.** The effect of changing constituents on tensile mechanical properties of HfNbTaTiZr high entropy alloy: A molecular dynamics study. *Heliyon* 2024, 10, e38350. 10.1016/j.heliyon.2024.e38350.
20. Uddin, M. M.; Hossain, M. Z.; **Kabir, M. H.**; Ghosh, S.; Haque, M. H.; Hossain, M. M.; Ali, M. A.; Akhtaruzzaman, M.; Alanazi, A. M.; Mondal, S.; Chowdhury, J.; Karmakar, D.; Jana, D. Synergistic effects of rare-earth ions (Ho, Yb) doping on the photo-catalytic efficacy of V₂O₅ for removal of pollutants from industrial wastewater. *Heliyon* 2024, 10, e37689. 10.1016/j.heliyon.2024.e37689.
21. Lozynskyi, V.; Trembach, B.; Hossain, M. M.; **Kabir, M. H.**; Silchenko, Y.; Krbata, M.; Sadovyi, K.; Kolomiitse, O.; Ropyak, L. Prediction of phase composition and mechanical properties Fe–Cr–C–B–Ti–Cu hardfacing alloys: Modeling and experimental validations. *Heliyon* 2024, 10, e25199. 10.1016/j.heliyon.2024.e25199.
22. Salam, M. A. R. B. A.; Rahman, M. A.; **Kabir, M. H.**; Alvarado, E. V.; Sadman, T.; Mahamud, R.; Cano, L.; Ashraf, A. Testing and modeling of an in situ shear exfoliated 2D nanocomposite coating casing material for the suppression of Li-ion battery fires in electric vehicles. *MRS Advances* 2023, 8, 953–959. 10.1557/s43580-023-00602-5.

Review articles

23. Shah, R.; Marussich, K.; **Kabir, M. H.**; Liang, H. Tribological Applications of Recycled and Waste Materials: A Review of Recent Advances and Future Directions. *Lubricants* 2026, 14, 104. 10.3390/lubricants14030104.
24. Shah, R.; Das, M.; **Kabir, M. H.**; Liang, H.; Salvi, L.; Bunting, T. L. Additive and Base Oil Trends in Electric Vehicle Applications. *Journal of Tribology* 2025, 147, 051106. 10.1115/1.4066650.
25. Dias, D.†; Bons, J.†; Kumar, A.†; **Kabir, M. H.**†; Liang, H. Forever Chemicals, Per-and Polyfluoroalkyl Substances (PFAS), in Lubrication. *Lubricants* 2024, 12, 114. 10.3390/lubricants12040114. [Equal contribution among the first four authors]
26. Uddin, M. M.; **Kabir, M. H.**; Ali, M. A.; Hossain, M. M.; Khandaker, M. U.; Mandal, S.; Arifutzzaman, A.; Jana, D. Graphene-like emerging 2D materials: recent progress, challenges and future outlook. *RSC Advances* 2023, 13, 33336–33375. 10.1039/d3ra04456d.

Conference proceedings

27. **Kabir, M. H.**; Ibrahim, H.; Billah, M. M. Effect of stabilizer on sol ageing for CuO thin films synthesized by sol-gel spin coating technique. *AIP Conference Proceedings* 2021, 2324, 030007. 10.1063/5.0037501.